

Mazidul Hasan

Senior QA Automation Engineer | AI Quality Engineer

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SUMMARY

Senior QA Automation Engineer with 6.5+ years of experience in test automation, quality engineering, and AI-driven QA systems. Specializes in designing scalable automation frameworks using Playwright, Selenium, Cypress, and API testing tools. Experienced in AI-powered testing including LLM evaluation, RAG validation, and agentic/multi-agent system testing. Strong background in CI/CD pipelines, JavaScript, and Python-based automation. Currently teaching AI in QA, prompt engineering, and LLM validation to QA professionals through a US-based training institute. Presently working with a US-based healthcare company, applying automation expertise to healthcare and billing systems

SKILLS

Core Automation Playwright, Selenium, Cypress, Rest Assured, Postman

Test Design & Management Test Plan, Test Case Design, Bug Tracking, Traceability, Shift Left

Testing Types Regression Testing, User Acceptance Testing (UAT), Feature Testing, Integration Testing, Performance Testing, Security Testing, Ad-hoc Testing, UI Testing

Programming JavaScript, TypeScript, Python, Java, SQL

Frameworks & Tools TestNG, JUnit, Cucumber, Maven, Gatling

AI & Automation LLM Evaluation & Testing, RAG Validation, Prompt Engineering for QA, AI Agent & Multi-Agent System Testing, AI-assisted Test Generation, Self-healing Automation, AI-based Test Data Generation, AI-driven API & UI Validation

DevOps & CI/CD Jenkins, Git, Docker, GitHub Actions

EXPERIENCE

Senior Software Engineer, QA

Therap (BD) Ltd.

Oct '19 — Present

Dhaka, Bangladesh

USA-based software company serving individuals with intellectual and developmental disabilities.

- Led Regression Testing, Feature Testing, Integration Testing, and User Acceptance Testing (UAT) planning and execution aligned with Agile sprints.
- Boosted API test efficiency by 75% by designing and implementing robust automation frameworks using Cypress and Playwright, significantly reducing test execution time.
- Enhanced UI test coverage by 25% by developing and executing comprehensive automation suites with Playwright, Selenium, TestNG, and Maven leading to more reliable releases.
- Optimized system performance by building a load-testing framework with Gatling, ensuring stability under high traffic conditions.
- Improved early defect detection by conducting in-depth requirement analysis to align testing strategies with project architecture and business goals.
- Elevated product quality by managing manual testing and defect tracking in JIRA, ensuring prompt resolution of issues and high-quality releases.
- Designed and tested AI-assisted automation workflows using Playwright and API integrations, enabling intelligent orchestration of end-to-end tests for dynamic AI-driven interfaces.
- Built AI-powered automation utilities for intelligent E2E and API validation, including LLM response validation and workflow verification to improve AI application reliability.
- Contributed to automation architecture for scalable testing of dynamic AI-integrated systems and agent-based testing workflows, focusing on reusability and maintainability.
- Developed reusable automation components for agent-based testing and implemented CLI-based automation utilities to streamline testing efficiency and debugging workflows.

Associate Software Engineer

OBHAI Solutions Ltd.

Mar '19 — Sep '19

Dhaka, Bangladesh

Ride-Sharing Company, Dedicated to revolutionizing urban mobility through innovative transportation solutions.

- Analyzed market trends and user feedback to identify opportunities for new feature development, leading to a 25% increase in user engagement.
- Coordinated with the team members to create and modify the app in a timely manner.

TEACHING & MENTORSHIP

QA Automation Instructor (Part-Time)

USA-Based Training Institute

- Deliver training on AI-powered software testing, quality engineering, and modern test automation practices.
- Teach prompt engineering for test generation, defect analysis, root-cause investigation, and automation acceleration.

- Train professionals on LLM evaluation, RAG validation, AI application testing strategies, and AI quality assurance methodologies.
- Guide students in designing and implementing AI-assisted QA automation frameworks using Playwright, Python, JavaScript, and modern LLM tools.
- Teach testing strategies for AI agents, agentic workflows, and multi-agent systems, including validation, observability, reliability, and failure analysis.
- Mentor QA engineers transitioning into AI-focused quality engineering and AI test automation roles.

EDUCATION

BSc in Computer Science and Engineering, BRAC University

Dec '18
Dhaka, Bangladesh

PROJECTS

AI Test Intelligence & Multi-Agent QA System [Link](#)

- Designed a multi-agent QA system with orchestrated AI agents for test generation, automation code creation, and test healing
- Implemented agentic workflow for separating roles: test planner, test generator, and self-healing executor
- Built intelligent decision layer to refine test cases using LLM reasoning loops
- Enabled adaptive QA automation for evolving UI and API changes

AI QA Copilot & Automation Studio Tools [Link](#)

- Built Chrome extension to record user interactions and convert them into Playwright, Selenium, and Cypress automation scripts
- Integrated LLM-based engine for converting manual test flows into structured test cases
- Added AI-assisted test case refinement and optimization layer
- Enabled hybrid manual + automated testing workflow generation

Agentic Jira-to-Playwright QA Automation System [Link](#)

- Built an agentic QA workflow integrating Jira, Atlassian MCP, Playwright, and LLM-powered agents for end-to-end test lifecycle automation.
- Implemented AI agents for Jira story analysis, test planning, Playwright automation code generation, and test execution.
- Automated defect detection and Jira bug creation with relevant test evidence and failure details.
- Enabled automated Jira status and comment updates after defect resolution.
- Connected requirements → test planning → automation → execution → defect management into a unified QA workflow.

AI Test Generation & API Intelligence Platform [Link](#)

- Built AI-powered platform that ingests Postman collections, API environments, and documentation to generate complete test suites
- Generates functional, negative, edge-case, and integration test cases using LLM reasoning
- Produces structured test results with AI-driven analysis and validation
- Bridges API documentation → test automation lifecycle

Test Data Intelligence Engine (NPM Package) [Link](#)

- Developed reusable test data generation library for healthcare domain
- Generates synthetic patient, medical, and structured healthcare datasets for automation testing
- Supports deterministic and randomized data generation for test reproducibility
- Used for API, UI, and performance testing scenarios

Core Automation Frameworks (Foundation Layer)

- [Playwright Boilerplate](#) (POM + CI + parallel execution)
- [Playwright API Automation](#) (fixtures, auth, schema validation)
- [RestAssured API Framework](#) (Java-based data-driven testing)